AMRITA VISHWA VIDYAPEETHAM
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COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-AOH-I011*

Team Name: *HACK ORBIT*

Domain: *Digital Technologies, AI & Cyber Security*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

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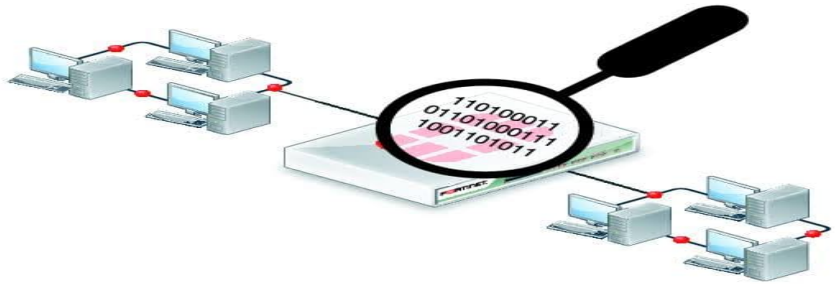
Presentation Outline

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Problem Statement

Basic Network Packet Sniffer & Analyzer

Everything we do online is chopped into tiny packets. Today it is hard for beginners to see them. We want a free tool that catches those packets and shows who is talking to whom, in plain English — like a CCTV camera for your Wi-Fi.



Problem Understanding

- **Who experiences this problem?** Network admins, security analysts, students and developers — and also ISPs, teachers and anyone whose video call has ever frozen.
- **What causes the problem?** Packets are a wall of numbers. Most tools are made for experts and show raw data, not simple answers, and capture, analysis and charts live in separate tools.
- **How does it affect them?** Fixes take longer, attacks slip by unnoticed, bandwidth gets wasted and learners never see how networks really work.

Existing Gaps & Challenges

- **Powerful tools are too complex:** Wireshark throws hundreds of fields at a beginner, tcpdump is just text with no charts, and Snort / Suricata take real expertise to set up.
- **Simple or paid options fall short:** ping / traceroute say something is wrong, not why. SolarWinds and PRTG are great but pricey and built for big firms.
- **Main challenge still unsolved:** nobody offers a free, beginner-friendly tool that captures, explains, charts and warns, all in one place.



Wireshark
Powerful, but scary for beginners



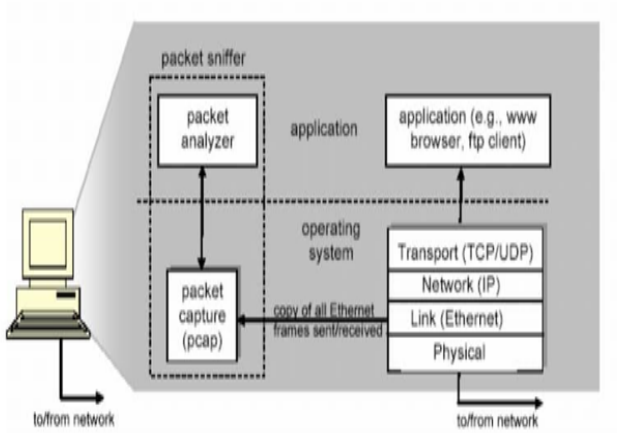
Snort / Suricata
Takes real expertise to set up



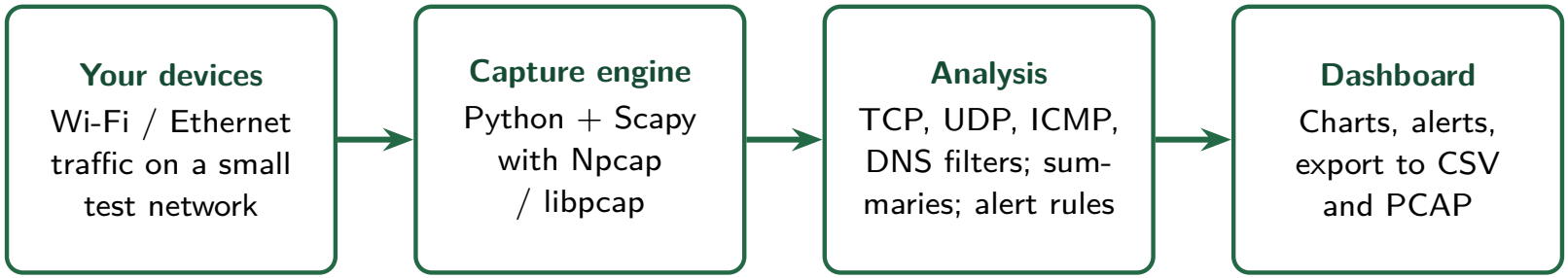
SolarWinds, PRTG
Great, but pricey and built for big firms

Our Proposed Solution

- **Core idea:** a network packet sniffer and analyzer that catches live traffic and shows it on a clean dashboard — who, where, which protocol, how big.
- **How it works:** live capture, then filters for TCP, UDP, ICMP and DNS, then summaries and charts, plus friendly alerts when it spots port scans or traffic spikes.
- **Main value to users:** beginners can actually read their traffic, slow or broken networks get fixed faster, and results export to CSV and PCAP.



Technical Approach



Packets are captured live, analysed and filtered, then shown on one dashboard with alerts and exports.

Tech Stack

Frontend: Dashboard with charts (Matplotlib)

Backend: Python + Scapy packet capture

Database: Pandas for analysis; CSV / PCAP export

AI/ML, APIs, or Hardware: Laptop (8 GB RAM), Npcap / libpcap

Innovation & Uniqueness

- **What is new or different?** One tool that captures, explains, charts and warns, instead of juggling several separate tools.
- **Why is it suited to the problem?** It shows traffic as simple answers, not raw data, so beginners, students and small teams can use it on a regular laptop.
- **Key differentiator:** Free and beginner-friendly, with built-in alerts for port scans and spikes — easy to use after one demo.

Feasibility & Scalability

- **Feasibility:** A regular laptop with 8 GB RAM, free tools (Python, Scapy, Npcap / libpcap, Pandas, Matplotlib), a team of four and a faculty mentor. Wireshark is used to double-check our results.
- **Scalability:** Start on a small test network and a second device, then grow by adding more protocol filters and alert rules.
- **Risks:** We capture only on networks we are allowed to monitor, and use sample packet files for demos.

Cost for Product (Optional)

- **Development or hardware cost:** Free software (Python, Scapy, Npcap / libpcap, Pandas, Matplotlib) on a laptop the team already has.
- **Hosting or maintenance cost:** None expected — the tool runs on a local laptop.
- **Estimated total:** Minimal — no expensive software needed.

Use Cases (compulsorily 3 use cases must be there)

- **Use case 1: Network admin.** Wi-Fi acts up — the admin sees which device and app is eating the bandwidth and fixes it quickly.
- **Use case 2: Security analyst.** Port scans and traffic spikes are flagged by friendly alerts, so suspicious activity is caught early.
- **Use case 3: Student or teacher.** Watch a network work live in the lab, with protocol charts and sample packet files, instead of just reading about it.

Impacts and Benefits

- **Benefit to target users:** Anyone can read their network traffic; problems get fixed faster and threats get caught earlier.
- **Social and economic impact:** No expensive software needed, and learners pick up real networking and security skills.
- **Expected measurable outcome:** Filters and charts work, alerts fire, data exports to CSV and PCAP, and it beats doing it by hand.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 4	Target 4.4	Quality Education	Students learn networking and security by seeing live traffic, building practical technical skills.
SDG 9	Target 9.1	Industry, Innovation and Infrastructure	Helps keep networks reliable and secure by catching slowdowns and threats early.

Individual Member Contribution

Member	Contribution or responsibility
Kavish Rahav D P	**Team Lead & Backend Developer** — builds the engine that catches packets and ties everything together
Karthikeyan K	**Network & Security Analyst** — works out what the traffic means and sets the alert rules
Ratchitha Kathiravan Vedhavalli	**Frontend / UI Developer** — designs the dashboard you actually see and use
Jeyanthan S	**Tester & Documentation Lead** — tests everything, checks it against Wireshark, writes it all up

Yukti ID

IC201810638